



University of
Nottingham

UK | CHINA | MALAYSIA

A large, high-resolution image of the Earth as seen from space, showing the Western Hemisphere with North and South America. The Earth is framed by a thin white rectangular border. The background is a deep black space filled with numerous small, distant stars.

COMP-2005
**Introduction to
Image Processing**

Lecture 3B
Non-Linear Filters



Learning Outcomes

IDENTIFY

1. Median Filtering
2. Anisotropic Diffusion
3. Bilateral Filtering





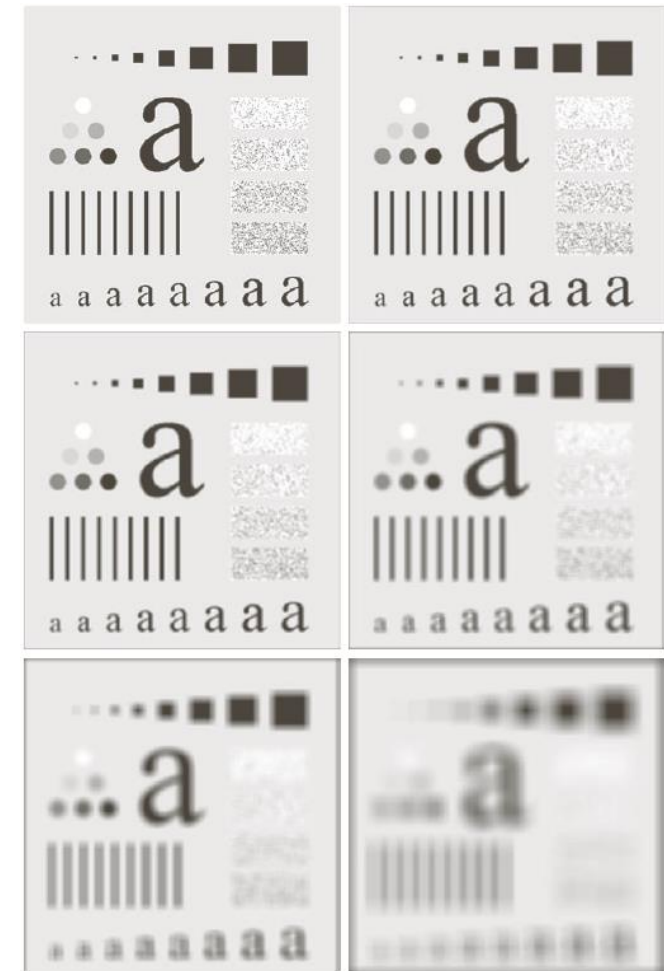
Non-Linear Filters

- Convolution with a mask of weights compute a linear function of a set of pixel values
- Many operations can be implemented this way, but not all:

- *Median filtering*
- *Anisotropic diffusion/Bilateral filtering*

Linear filters smooth sharp image changes, **nonlinear filters** tend to preserve or even enhance them

Difference



Gaussian Smoothing



University of
Nottingham

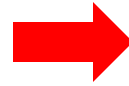
UK | CHINA | MALAYSIA

Median Filtering



Salt and Pepper Noise

Sometimes sensors either fail to respond or saturate in error



- A **false saturation** gives a **white spot** in the image (**salt**)
- A **failed response** gives a **black spot** in the image (**pepper**)
- Sometimes called **speckle noise**



1%



10%



20%

An image with varying amounts of salt and pepper noise added



Reducing Salt and Pepper Noise



Original



Salt & Pepper



Original



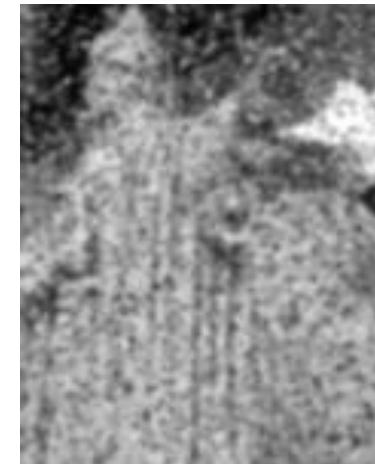
Salt & Pepper



Mean Filtering



Gaussian Filtering





The Median Filter

Alternative: Median Filter

- Statistically the median is the middle value in a set
- Each pixel is set to the median value in a local window
- Result is a real pixel value, not a combination
- Noise pixels are outliers
- Noise would have to affect $>1/2$ the pixels to appear in the output

123	124	125
129	127	9
126	123	131

123	124	125	129	127	9	126	123	131
-----	-----	-----	-----	-----	---	-----	-----	-----

Find the values in a local window

9	123	123	124	125	126	127	129	131
---	-----	-----	-----	-----	-----	-----	-----	-----

Sort them

9	123	123	124	125	126	127	129	131
---	-----	-----	-----	-----	-----	-----	-----	-----

Pick the middle one

A mean filter would give 113



The Median Filter



Original



Salt & Pepper



Gaussian



ACK: Prof. Tony Pridmore, UNUK



University of
Nottingham

UK | CHINA | MALAYSIA

Anisotropic Diffusion



Anisotropic Diffusion

- Median filtering is good given small regions of speckle noise, less good if edges are important
- There exist explicit edge-preserving smoothing ops

Diffusion

- Spreading out
- Mean and Gaussian filters can be seen as diffusion processes

Anisotropic

Not the same in all directions

Basic IDEA

- Mean and Gaussian filters make each pixel more like its neighbours
- **Anisotropic diffusion** makes each pixel more like those neighbours that it is already similar to



Anisotropic Diffusion

We have a similar function, $s(p,q)$

- $s(p,q)$ has values in the range from 0 to 1
- If the pixels p and q are **similar** then $s(p,q)$ is close to **1**
- If the pixel p and q are **different** then $s(p,q)$ is close to **0**

We use $s(p,q)$ to compute a weighted average of pixel values

- The new value at a pixel p , is based on all its neighbours, q

$$p' = \frac{\sum q \times s(p,q)}{\sum s(p,q)}$$



The Similarity Function

- The smoothing function, $s(p, q)$ needs to be found
- If d is the difference between p and q and D is the maximum possible difference we can use:

$$\frac{D - d}{D}$$

$$s(p, q) = e^{\left(\frac{p - q}{K}\right)^2}$$

Other functions often used include:

$$s(p, q) = \frac{1}{1 + \left(\frac{p - q}{K}\right)^2}$$

K determines the amount of smoothing



Anisotropic Diffusion

The examples here used the similarity function:

$$s(p,q) = \frac{1}{1 + \left(\frac{p - q}{K} \right)^2}$$

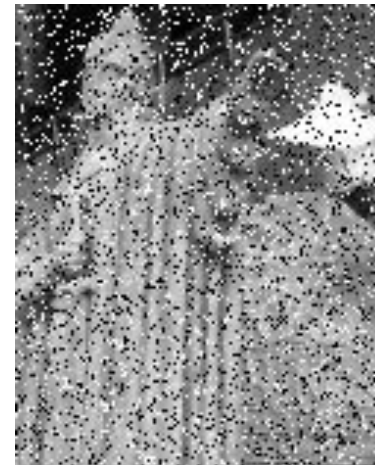
With $K = 25$



Salt & Pepper



Gaussian





Anisotropic Diffusion

A higher value of **K** gives greater smoothing, but **edges are still (quite) sharp**



$K = 5$



$K = 25$



$K = 50$



$K = 100$

ACK: Prof. Tony Pridmore, UNUK



Anisotropic Diffusion

We can apply the filter repeatedly to give greater smoothing



Original



1 Iteration



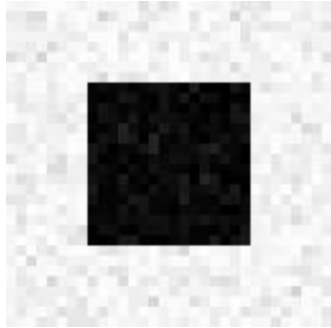
2 Iteration



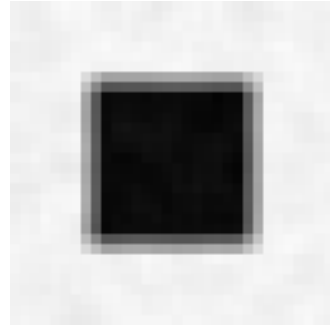
3 Iteration



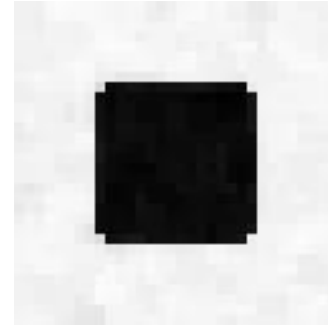
Reducing Noise near Edges



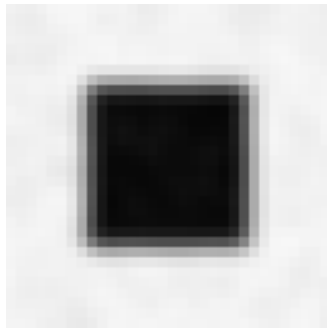
Noise Image



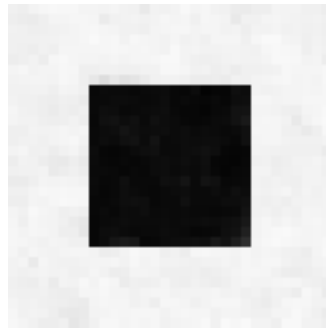
Mean



Median



Gaussian



Anisotropic Diffusion

Anisotropic diffusion
is to mean filtering
as **????** is to
Gaussian filtering...



University of
Nottingham

UK | CHINA | MALAYSIA

Bilateral Filtering



Bilateral Filtering

- Anisotropic Diffusion is related to mean filtering
- If the similarity function is always 1 we get a mean filter

$$p' = \frac{\sum q \times s(p, q)}{\sum s(p, q)}$$

Sums pixel values in a region

Counts pixel values in a region

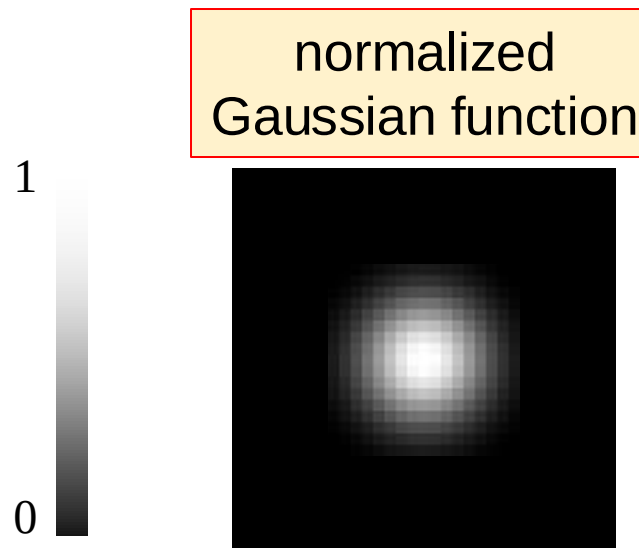
Bilateral filters modify Gaussian smoothing in a similar way

- One Gaussian weights pixels that are near the source
- Another Gaussian weights pixels that have similar intensity to the source pixel



Gaussian Smoothing Again

$$GB[I]_{\mathbf{p}} = \sum_{\mathbf{q} \in S} G_{\sigma}(\|\mathbf{p} - \mathbf{q}\|) I_{\mathbf{q}}$$





Bilateral Filtering

$$BF[I]_p = \overset{\text{new}}{\frac{1}{W_p}} \sum_{q \in S} \overset{\text{not new}}{G_{\sigma_s}(\|\mathbf{p} - \mathbf{q}\|)} \overset{\text{new}}{G_{\sigma_r}(|I_p - I_q|)} I_q$$

Diagram illustrating the components of the Bilateral Filtering equation:

- Normalization factor:** $\frac{1}{W_p}$ (pink box)
- Space weight:** $G_{\sigma_s}(\|\mathbf{p} - \mathbf{q}\|)$ (orange box, labeled "not new"). Below it is a 2D Gaussian kernel visualization.
- Range weight:** $G_{\sigma_r}(|I_p - I_q|)$ (blue box, labeled "new"). Below it is a 1D Gaussian kernel visualization along the intensity axis I .

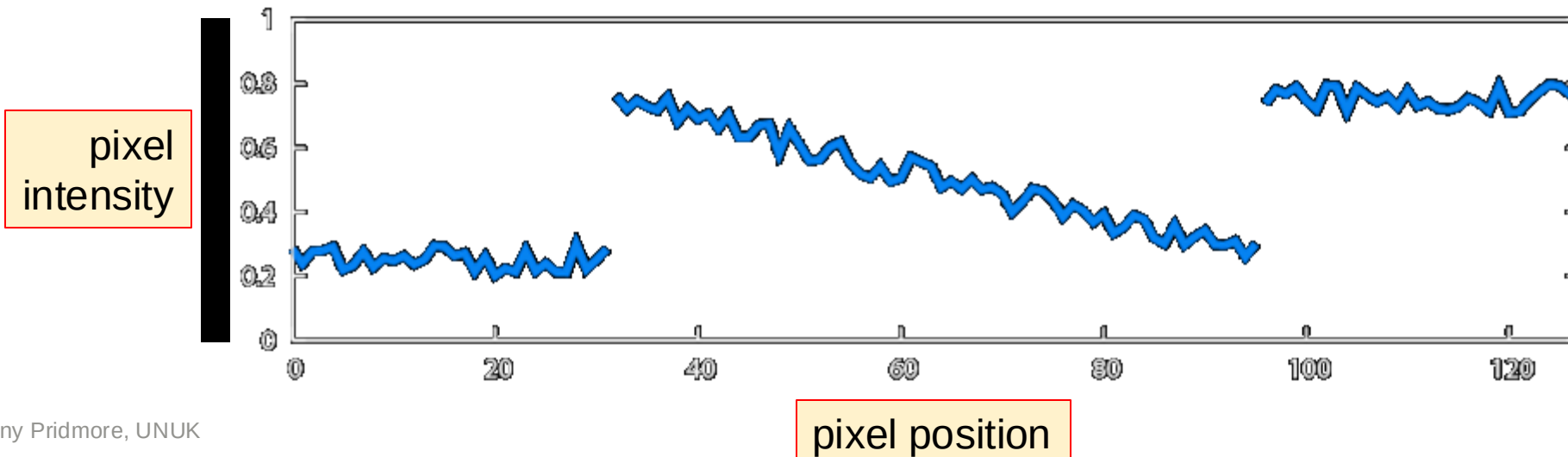


A One Dimensional Example

1D image = line of pixels



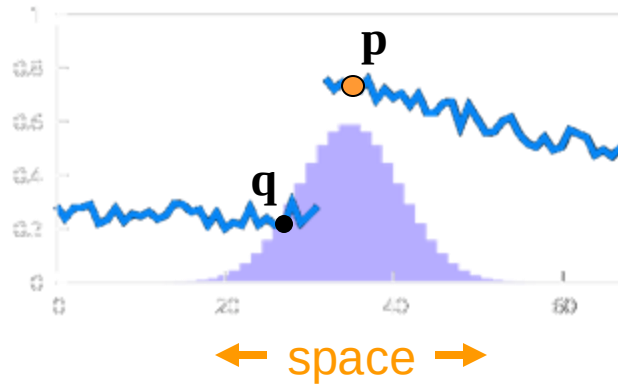
Better visualised as a plot



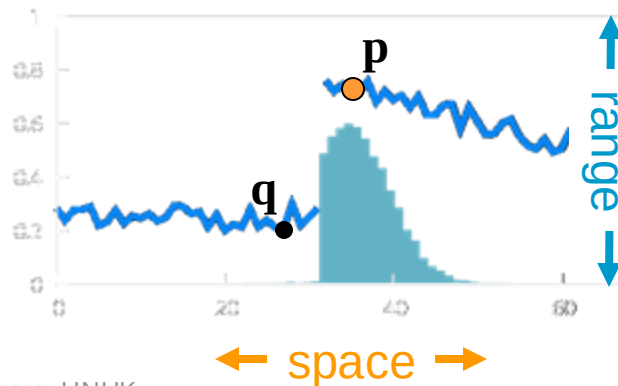


Gaussian & Bilateral Smoothing

Gaussian blur



Bilateral filter



$$GB[I]_p = \sum_{q \in S} G_{\sigma}(\|\mathbf{p} - \mathbf{q}\|) I_q$$

space

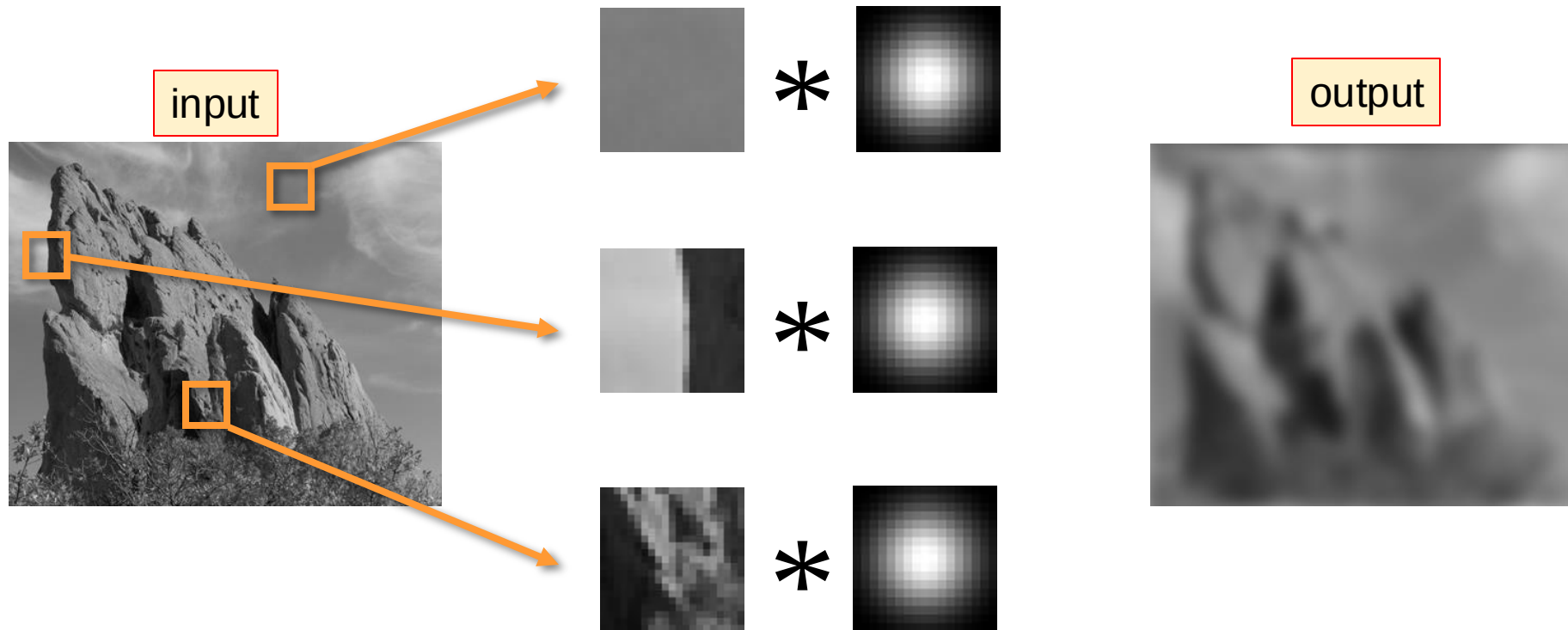
$$BF[I]_p = \frac{1}{W_p} \sum_{q \in S} G_{\sigma_s}(\|\mathbf{p} - \mathbf{q}\|) G_{\sigma_r}(|I_p - I_q|) I_q$$

normalization space range

ACK: Prof. Tony Pridmore, UNUK

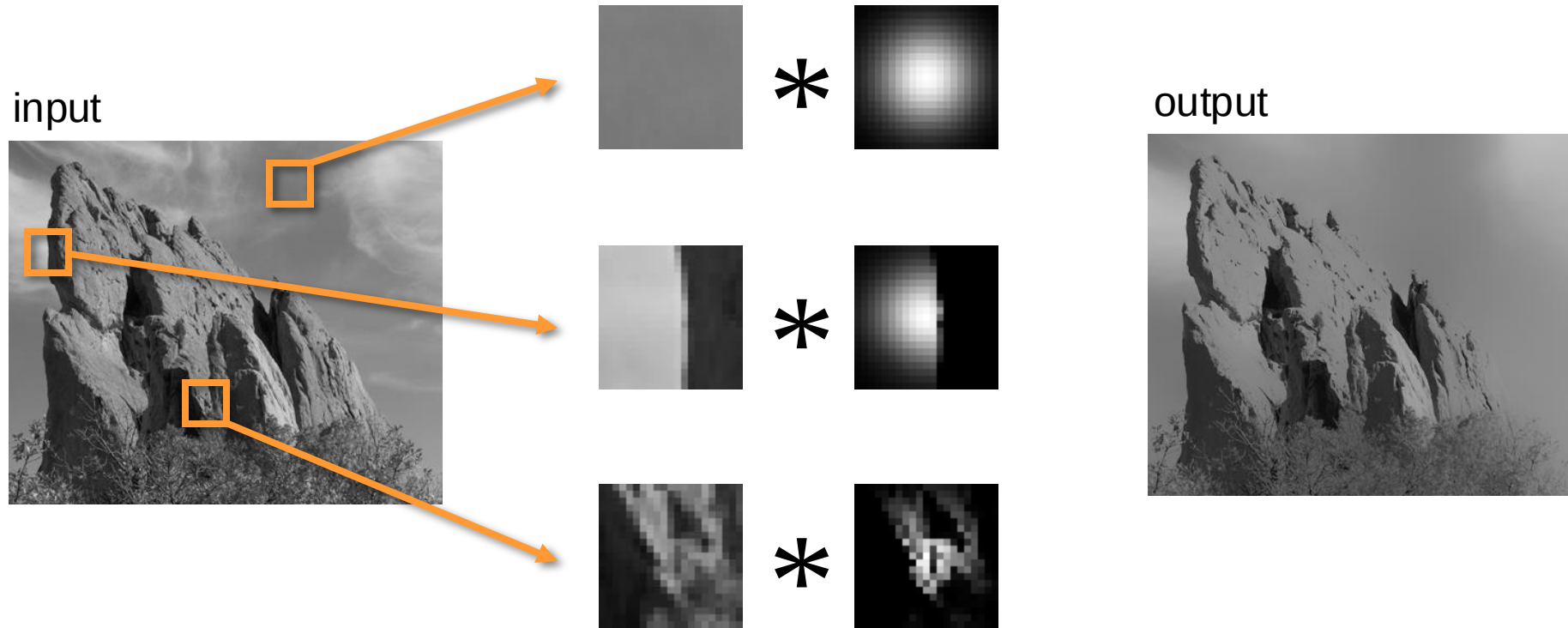


In 2D: Gaussian Smoothing





In 2D: Bilateral Filtering



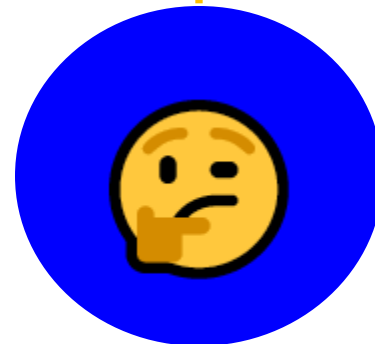
The kernel shape depends on the image content.



Summary



1. Median Filtering
2. Anisotropic Diffusion
3. Bilateral Filtering





University of
Nottingham

UK | CHINA | MALAYSIA

Questions



University of
Nottingham

UK | CHINA | MALAYSIA

NEXT:

**Thresholding
&
Binary Images**